

- 7 A program is being developed to implement a customer loyalty scheme for a coffee shop.

The programmer has decided that the following data items need to be stored for each customer:

Data item	Description
customer ID	a unique six-digit string
points	an integer value that is increased by one for every cup of coffee the customer orders

When a customer visits the shop and orders coffee, the scheme operates as follows:

- The total number of points is increased by the number of coffees ordered.
- If just one cup of coffee is ordered and the number of points goes above 10, then:
 - the cup of coffee they have just ordered is given to them free of charge
 - the number of points is reduced by 11.
- If the order is for multiple coffees and the number of points goes above 10, then:
 - they get one coffee free of charge for every 11 points
 - the number of points is reduced by 11 for each free coffee.

For example, the:

- customer currently has 9 points
- customer orders 16 cups of coffee
- total number of points now becomes 25
- customer gets 2 free coffees and now has 3 points left.

The programmer has defined a program module that is called every time a customer places an order:

Module	Description
<code>CustomerOrder()</code>	• called with two integer parameters: ◦ the number of coffees ordered ◦ the current number of points • output a suitable message giving the number of free coffees • return a value for the new points total.

